

Day 1: Understanding Programming + First Code

Hour 1 – Think Like a Developer

What is Programming?

Programming = Giving instructions to a computer to perform tasks.

Think like this:

→ You →  Chef

→ Computer →  Obedient but very dumb assistant

✓ Clear instructions → Perfect result

✗ Unclear instructions → Error

 A cute robot chef cooking in a futuristic kitchen

What is Python?

Python is:

- ✓ Easy to read
- ✓ Beginner friendly
- ✓ Very powerful

Used in:

- Web Development
- AI / Machine Learning
- Automation
- Data Science
- Backend Development

Companies Using Python



Google



Instagram



Netflix



Spotify

Python is powerful but simple. That's why we're learning it.

Installation Guide

1. Install Python

Go to: python.org and download the latest version.

- ✓ Check "Add Python to PATH" during installation.

Verify installation in Command Prompt:

```
python --version
```

If version shows → Done 

2. Install VS Code

Download from: code.visualstudio.com

Install Extensions inside VS Code:

- ✓ Python (Official by Microsoft)
- ✓ Code Runner (Optional but helpful)

Hour 2 – First Python Program

Create a new file named:

```
day1.py
```

Write the following code:

```
print("Hello World")
```

- ✓ Run the file in VS Code.
- ✓ See the magic happen in the terminal.

This is your first step toward becoming job-ready.



Understanding Variables

Variable = Container that stores data.

Example in Python:

```
name = "Saurav" age = 20 height = 5.8 is_student = True
```

- Variables are created the moment you first assign a value to them.
- You don't need to declare their type beforehand.

Core Data Types

- **int** → Whole numbers (e.g., 10)
- **float** → Decimals (e.g., 10.5)
- **string** → Text (e.g., "Hello")
- **boolean** → Logic (e.g., True / False)

How to check a variable's type:

```
# This will output: print(type(age))
```


Practice Tasks



Print Details

Print the following exact sentence using Python:

"My name is Saurav and I am 20 years old."



Age Calculator

Write a script to calculate your current age using your birth year and the current year.



Simple Math

Make a simple calculator:

```
a = 10 b = 5  
print("Addition:", a + b)
```




Mini Challenge

Create a program that prints:

"Welcome to Saurav's Future Tech Company"

Then, print out your:

- ✓ Your Goal
- ✓ Your Dream Salary
- ✓ Your Target Job Role

Mindset Lesson

“

Don't rush. Day 1 is about clarity, not speed. Strong foundation = Easy future learning.

”

When finished, say:

🔥 "Day 1 Completed"

DAY 2 – Operators + User Input

Goal: 2 Hours

🎯 Make interactive programs using logic.

Today your program becomes powerful.

Arithmetic Operators

Operator	Name	Example (a=10, b=3)
+	Addition	$a + b = 13$
-	Subtraction	$a - b = 7$
*	Multiplication	$a * b = 30$
/	Division	$a / b = 3.33$
//	Floor Division	$a // b = 3$
%	Modulus	$a \% b = 1$
**	Power	$a ** b = 1000$

```
a = 10 b = 3 print(a + b) # 13
print(a % b) # 1 print(a ** b) # 1000
```

⚠️ Understand % (Modulus) very carefully. It returns the remainder.

Comparison & Logical Operators

Comparison Operators

Return True or False.

<code>==</code> Equal	<code>!=</code> Not Equal
<code>></code> Greater	<code><</code> Less
<code>>=</code> Greater/Equal	<code><=</code> Less/Equal

```
print(10 > 5) # True
```

Logical Operators

→ and

→ or

→ not

```
age = 20 has_id = True print(age >= 18 and has_id)
```

This is how login systems work!

User Input

Allow users to interact with your program:

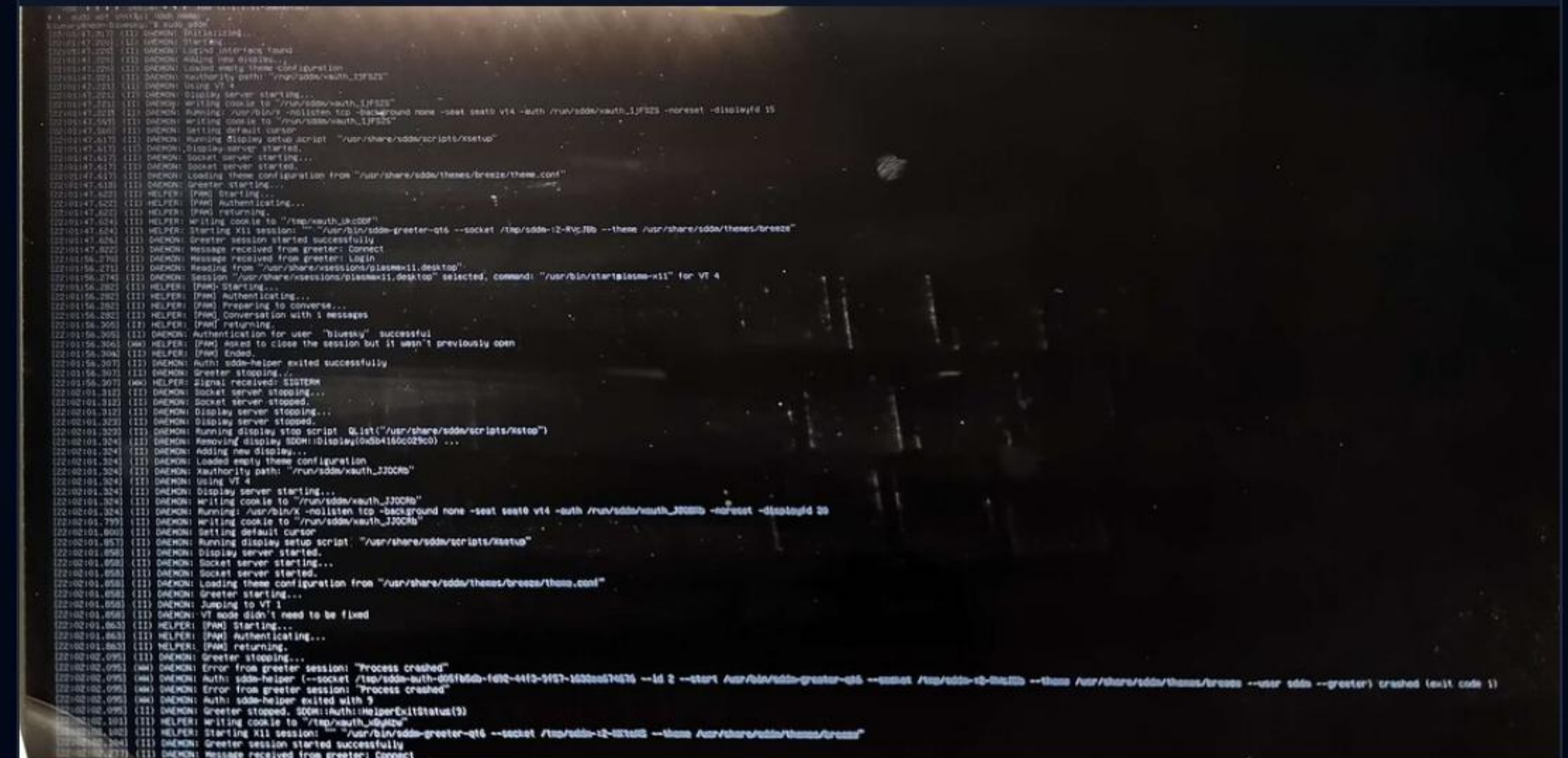
```
name = input("Enter your name: ")
print("Hello", name)
```

⚠ Important Warning

Input always comes as a string.

For numbers, you must convert it:

```
age = int(input("Enter your age: "))
```



Practice Tasks



Basic Calculator

Create a calculator that takes user input for two numbers and adds them together.



Area of Rectangle

Ask user for length and width. Calculate and print the area.



Even or Odd Checker

Hint: Use the modulus operator.
`if number % 2 == 0:`



Simple Interest

Formula: $SI = (P * R * T) / 100$



Mini Challenge – Pass/Fail System

Basic Logic

Ask the user for their marks and output the result.

→ If marks ≥ 40 → Pass

→ Else → Fail

This is your first real logic system!

Advanced Version

Implement grading logic:

Marks Range	Grade
80+	Distinction
60 – 79	First Class
40 – 59	Pass
Below 40	Fail

Moment of Truth

Did you actually type these programs
or just read them? 😊

If you successfully wrote the code, reply with:



"Day 2 Completed"

Image Sources



https://image.slidesdocs.com/responsive-images/background/futuristic-cooking-kitchen-scene-with-3d-rendered-robot-chef-powerpoint-background_c0647f7f04_960_540.jpg

Source: slidesdocs.com



<https://images5.alphacoders.com/134/thumb-1920-1349437.png>

Source: wall.alphacoders.com



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